Road damage. New country.

Can a model trained elsewhere detect small or thin road damage?

01 / THE SHARED TEST

Different roads. Different cameras. Same task.

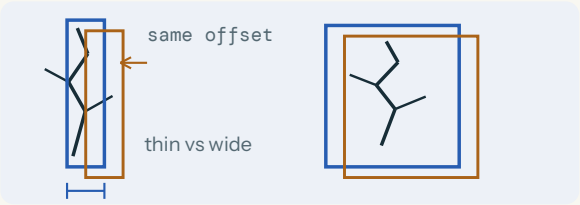
TRAIN	FREEZE	TEST
3 source countries	Model + settings	1 unseen country

Rotate the test country four times. No target-country data for training or tuning.

RDD2022 / Japan, India, United States, Czech Republic / YOLO11

02 / THREE SEPARATE RESEARCH QUESTIONS

A Match thin boxes differently

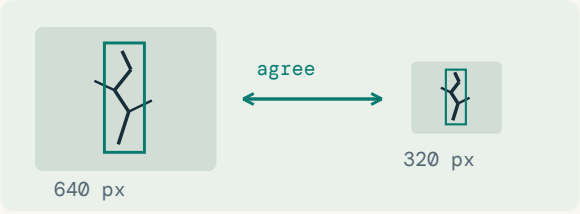


Blue: label box / Amber: prediction

Measure matching error relative to a box's **short side**. Change which predictions learn from it, and their weights.

TRAINS	Modified YOLO
CONTROL	Equally tuned plain NWD
MEASURE	Detection accuracy (mAP)

B Learn across image scales

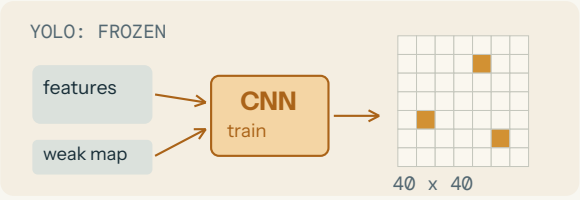


Extra views during training; one view at test time.

Use two scales of the same image. Add a training loss that encourages agreement on **visible small or thin damage**.

TRAINS	YOLO + consistency loss
CONTROL	Same loss on all eligible boxes
MEASURE	Recall + false positives

C Warn about likely misses



A warning grid, not recovered cracks.

Keep YOLO fixed. Train a small **CNN** on its internal features and weak detections to flag likely missed-box centres.

TRAINS	Separate CNN head
CONTROL	Learned weak-map-only head
MEASURE	Miss-location AUPRC

3 SEEDS / MATCHED CONTROLS / ABLATIONS / SOURCE-ONLY TUNING

All three train neural models. Improvements remain to be tested.

Based on proposal v8. Diagrams are schematic; no results are shown.